

Motion-enabled tomography via Gaussian mixture models

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Context & Problem Statement

- We consider recovering unknown properties of objects in motion using tomography. Performing such a task enables automation of geological inspection processes, but can also be used in applications for reconstructing objects with unknown orientation, such as in cryo-EM.
- Let $f: \mathbb{R}^d \times [0, t_{\max}] \rightarrow \mathbb{R}$ be the unknown function in motion. The goal is to recover $t \mapsto f(\cdot, t)$ on $\mathbb{R}^d$ from time series $t \mapsto g(\cdot, t)$, where $g(\ell, t) = \mathcal{X}(f(\cdot, t))(\ell)$ is the ray transform of $f(\cdot, t)$ on the line $\ell \in \mathbb{M}$ for some manifold $\mathbb{M}$ of lines in $\mathbb{R}^d$.

Simulated Data: $N = 5$ Particles

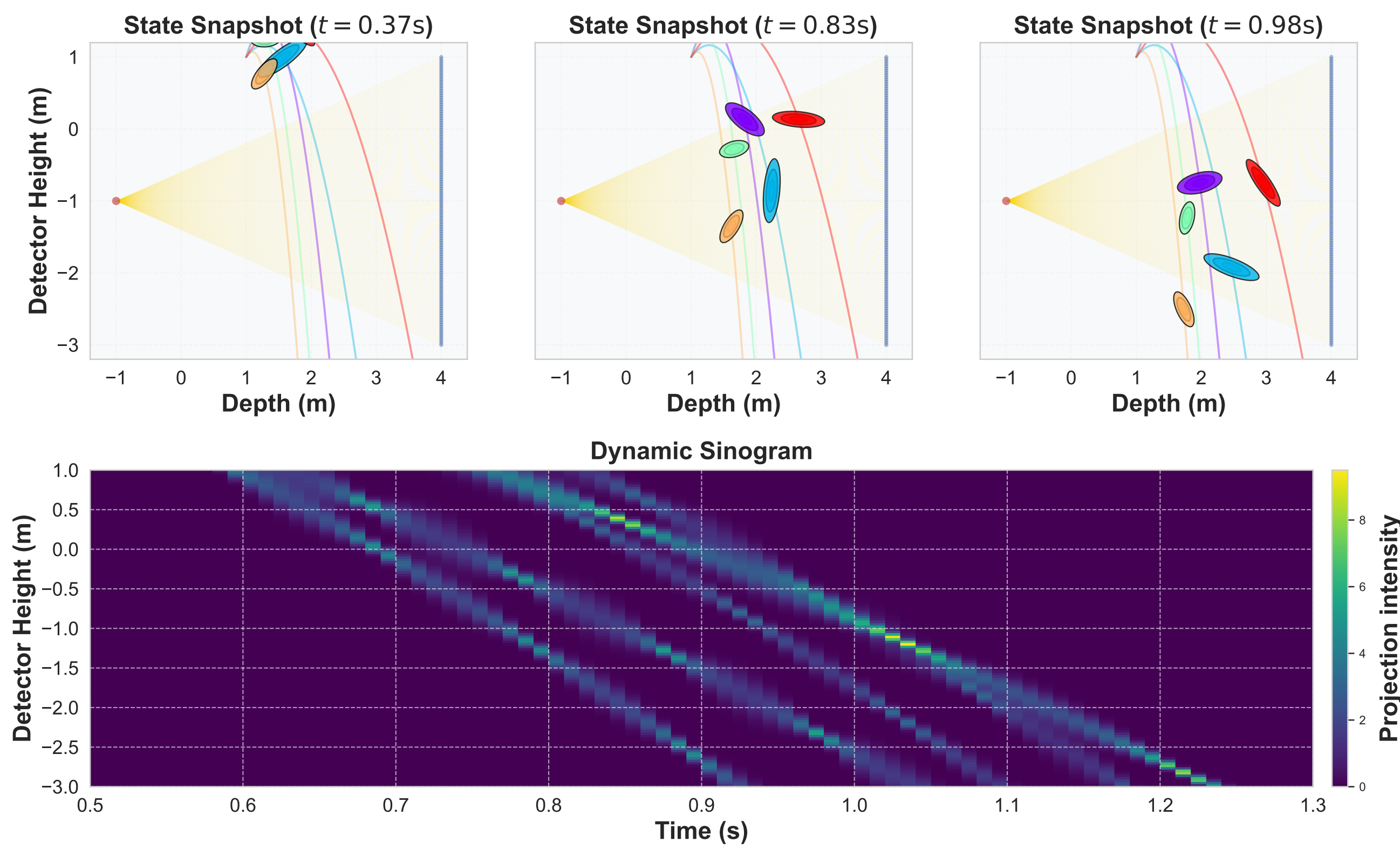


Figure 1: Top: The $N = 5$ unknown Gaussian particles that we seek to recover, based on their sinogram time series. Bottom: The complete 1D sinogram time series, plotted as a function of detector height at each observation time.

Modelling & Inversion

1. Spatio-Temporal State Model

Approximate f as the linear combination of N unnormalised Gaussian particles:

$$f(x, t) \approx \hat{f}[\Gamma](x, t) := \sum_{n=1}^N \beta_n(t) \cdot \mathcal{N}(x; C[\eta_n](t), M_n(t)), \quad (1)$$

where

- $C[\eta_n](t)$ encodes the parabolic trajectory of the n -th particle; parametrised by η_n .

- The matrix

$$M_n(t) = R_t \Theta_n U_n^\top U_n R_t^\top \Theta_n^\top$$

encodes the fused rotation and morphology of the n -th particle via the n -th angular velocity vector Θ_n and n -th skewness matrix U_n respectively.

- $\beta_n(t) = \alpha_n (2\pi)^{d/2} | \frac{1}{2} M_n(t) |^{-1}$, where $\alpha_n \in \mathbb{R}^+$ is an attenuation coefficient to be determined.
- The parameter Γ is the concatenated vector of all (η_n, U_n, Θ_n) across all n .

2. Forward Model (Ray Transform)

Assuming the lines $\ell \in \mathbb{M}$ are parametrised by source-detector pairs $(s, r) \in \mathbb{R}^d \times \mathbb{R}^d$, the forward model of the GMM (1) parametrised by Γ is

$$F[\Gamma](s, r, t) := \mathcal{X}(\hat{f}[\Gamma](x, t)) = \sum_{n=1}^N \mathcal{X}(\beta_n(t) \cdot \mathcal{N}(x; C[\eta_n](t), M_n(t))). \quad (2)$$

Since Gaussians are closed under $\mathcal{X}$, eq. (2) can be written exactly as a function of Γ .

3. The Inverse Problem

The sinogram data by which Γ is observed is denoted by $g[m]_{m \in \mathbb{I}}$, where $m = (m_s, m_r, m_t) \in \mathbb{N}^3$ is a multi-index that yields a tuple $(s_{m_s}, r_{m_r}, t_{m_t})$. The inverse problem is then that of computing a Γ^* such that

$$g[m] \approx F[\Gamma^*](s_{m_s}, r_{m_r}, t_{m_t}). \quad (3)$$

From Joint to Sequential Recovery of Motion & Morphology

- Least-squares type solutions of the inverse problem (3) are obtained by solving

$$\Gamma^* \in \arg \min_{\Gamma} L(\{g[m]\}_{m \in \mathbb{I}}, \{F[\Gamma](s_{m_s}, r_{m_r}, t_{m_t})\}_{m \in \mathbb{I}}). \quad (4)$$

The latent parameter Γ can be split into projectile, rotational and morphological components as $\Gamma = [\Gamma_P, \Gamma_R, \Gamma_M]$.

- Since the mode of the ray transform of a Gaussian is independent of its morphology, the minimisation problem can be decomposed into the two sub-problems

$$\Gamma_P^* \in \arg \min_{\Gamma_P} L_P(\{g[m]\}_{m \in \mathbb{I}}, \{F[\Gamma_P, \Gamma_R, \Gamma_M](s_{m_s}, r_{m_r}, t_{m_t})\}_{m \in \mathbb{I}}) \quad (5)$$

$$[\Gamma_R^*, \Gamma_M^*] \in \arg \min_{[\Gamma_R, \Gamma_M]} L_D(\{g[m]\}_{m \in \mathbb{I}}, \{F[\Gamma_P, \Gamma_R, \Gamma_M](s_{m_s}, r_{m_r}, t_{m_t})\}_{m \in \mathbb{I}}), \quad (6)$$

where L_P and L_D are task-specific loss functions. Solutions of (4) can then be obtained by instead solving the simpler (5) and (6).

Algorithm

Stage 1: Trajectory Estimation

1. Approximate the modes $\tilde{r}(t_{m_t})$ from the data using a peak detection method.
2. For estimates $\{\eta_n\}_{n=1}^N$, apply $\hat{r}[\eta_n](t_{m_t}) = s + \lambda_n(s - C[\eta_n](t_{m_t}))$, where λ_n is known.
3. Compute the loss of $\{\eta_n\}_{n=1}^N$ by optimally-assigning the observed $\tilde{r}(t_{m_t})$ to $\hat{r}[\eta_n](t_{m_t})$ at each t_{m_t} . Since $\hat{r}[\eta_n]$ is explicit and smooth in η_n , gradient descent can be utilised.

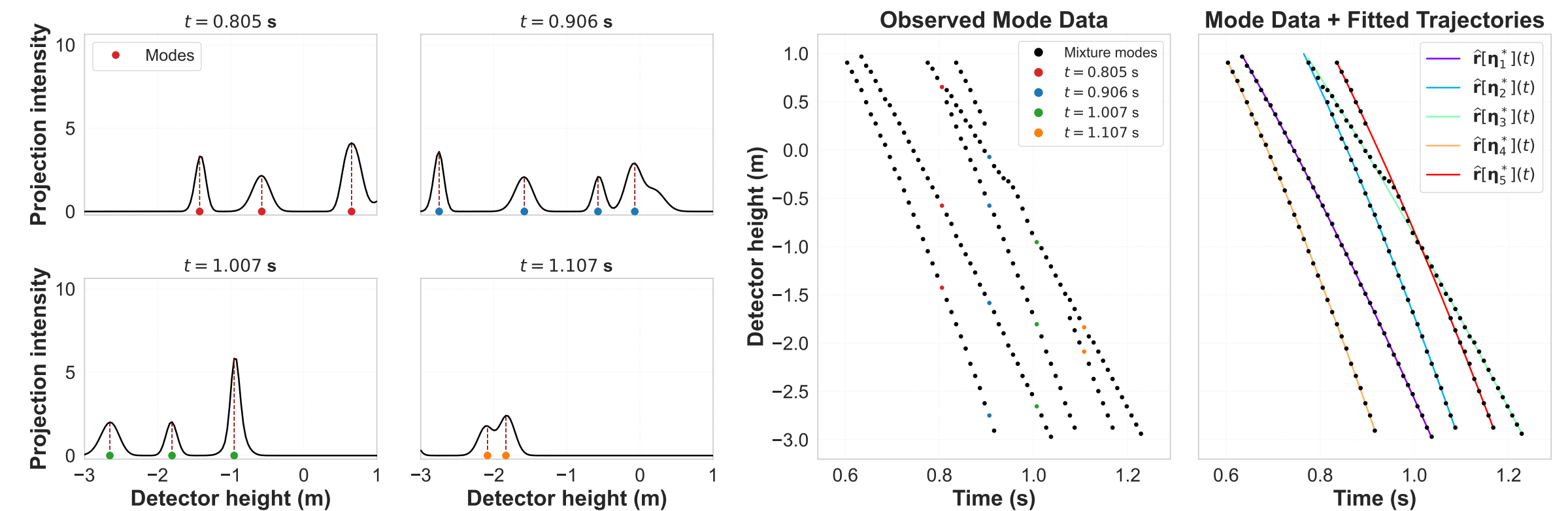


Figure 2: Left: Demonstration of computation of the modes at four selected observation times. Centre: The complete observed mode time series used for fitting. Right: The fitted trajectories that are the output of (5).

Stage 2: Rotation and Morphology Estimation

1. To initialise for rotation, use grid search to numerically approximate

$$\arg \min_{\Theta_n} \left\| g[m] - \sum_{\ell \neq n} F[\gamma_\ell](s, r_{m_r}, t_{m_t}) - F[\gamma_n(\Theta_n)](s, r_{m_r}, t_{m_t}) \right\|_{\ell^2}, \quad n = 1, \dots, N.$$

2. To initialise for attenuation, use Non-Linear Least Squares (NNLS) to approximate

$$\arg \min_{\alpha} \|\Phi \alpha - g[m]\|_{\ell^2}^2, \quad \alpha = (\alpha_1, \dots, \alpha_N)^\top.$$

3. Lastly, solve for (6), with L_D chosen as a smooth approximation to the L_1 Huber loss function.

Simulated Data with Reconstructed Solutions

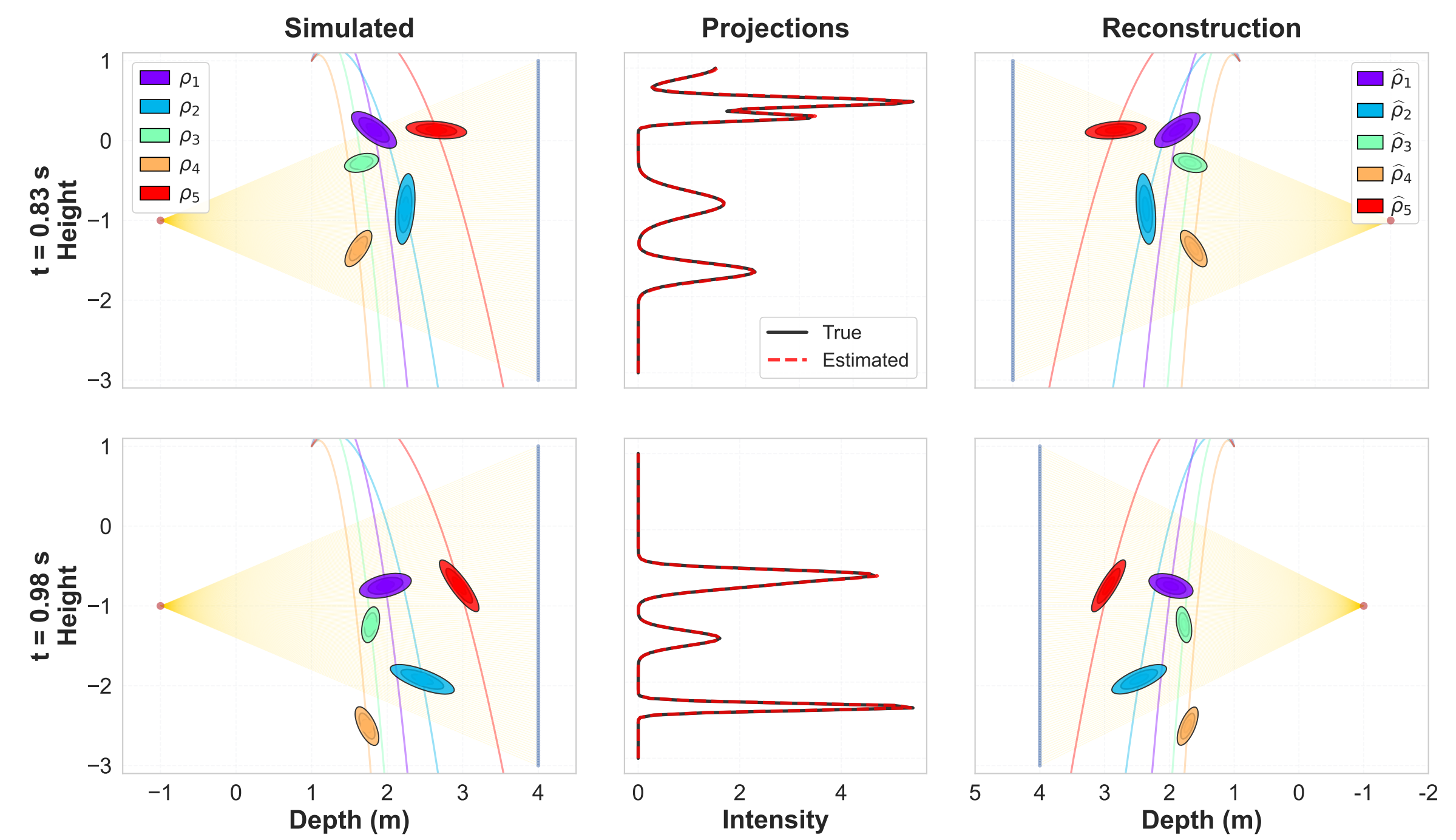


Figure 3: Left: The 2nd and 3rd panels from Figure 1. Right: The reconstructed solution at the same sample times. Centre: The corresponding sinograms that inform the reconstruction fit.

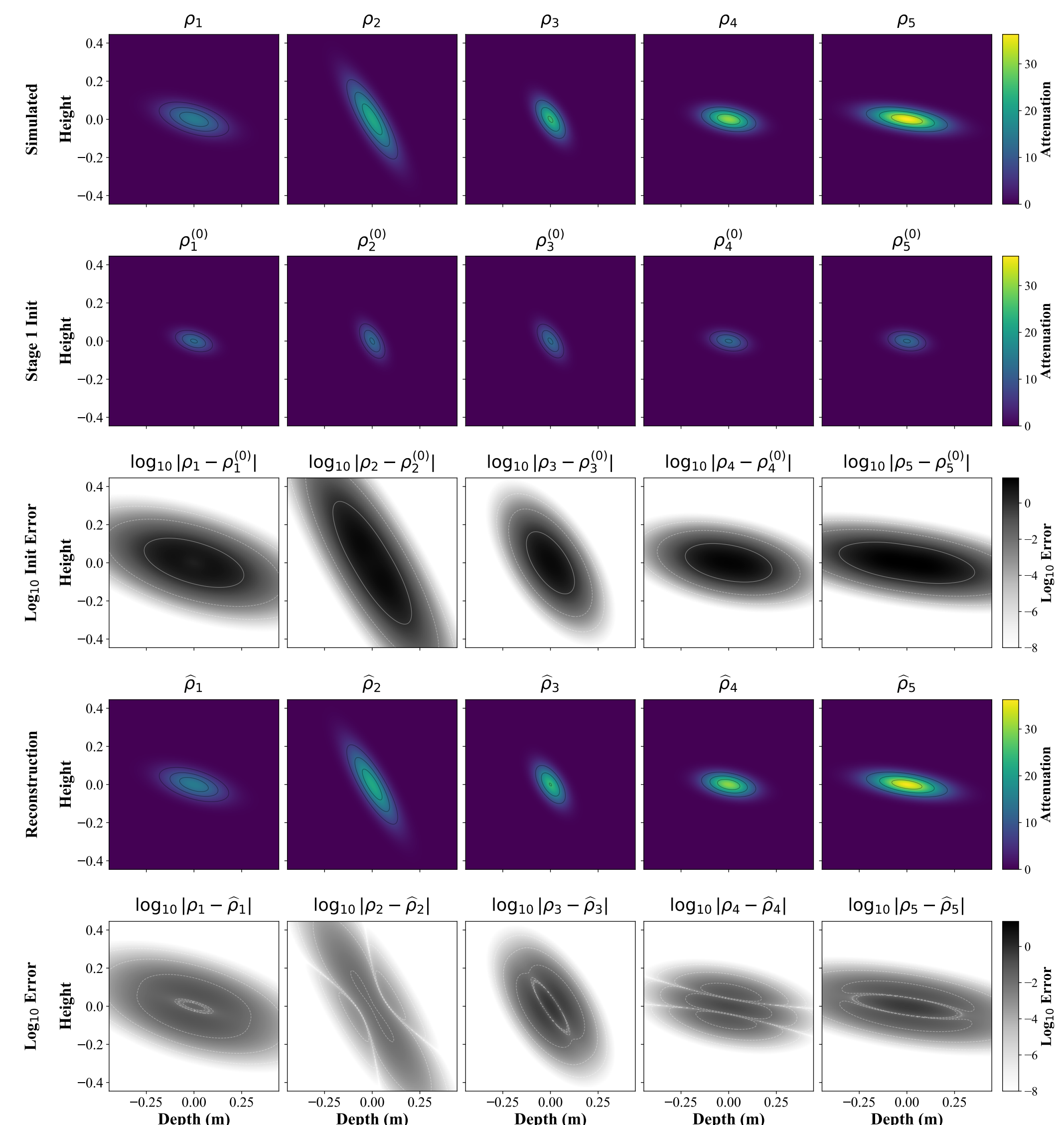


Figure 4: First row: the unknown, simulated particles. Second row: the initialized estimates, prior to Stage 1. Third row: The corresponding errors. Fourth row: The reconstructed solutions. Fifth row: The final errors between simulated and reconstruction. All particles are oriented using the rotational motion taken at $t = 0$.